

Exercise 1

A processor has 16-bit virtual addresses and a page size of 256 B. The system has a single level page table. The processor is equipped with a 2-entry fully associative TLB with LRU replacement policy. Moreover, it has a direct mapped L1 cache of 256 B with a block of 16 B. Consider the following assembly code and compute EMAT and AMAT. Assume that $T_{L1HIT} = 1$ cycle, Miss Penalty = 50 cycles, $T_{TLBHIT} = 1$ cycle, $T_{PT} = 20$ cycles (time taken by the page walk in case of TLB miss).

```
lw $t0, 0x1000($zero)
lw $t1, 0x1004($zero)
lw $t2, 0x1100($zero)
lw $t3, 0x1000($zero)
```

Consider that the page table has been initialized by the OS with the following content:

VPN	PFN	Valid
0x10	0x20	1
0x11	0x21	1

Solution

Virtual address split: 8 bit VPN and 8 bit offset

Physical address split: 8 bit tag, 4 bit cache index, 4 bit offset

TLB and cache initially empty.

Valid	VPN	PFN
0		
0		

Index	Validity bit	Tag	Block
0	0		
1	0		
...	...	...	...
15	0		

- lw \$t0, 0x1000(\$zero)

The virtual address 0x1000 is translated into the physical address 0x2000. There is a TLB miss and a L1 cache miss. After the lw instruction the status of the TLB and of the L1 cache is:

Valid	VPN	PFN
1	0x10	0x20
0		

Index	Validity bit	Tag	Block
0	1	0x20	16 B
1	0		
...	...	...	...
15	0		

- lw \$t0, 0x1004(\$zero)

The virtual address 0x1004 is translated into the physical address 0x2004. There is a TLB hit and a L1 cache hit. After the lw instruction the status of the TLB and of the L1 cache is:

Valid	VPN	PFN
1	0x10	0x20
0		

Index	Validity bit	Tag	Block
0	1	0x20	16 B
1	0		
...	...	...	...
15	0		

- lw \$t0, 0x1100(\$zero)

The virtual address 0x1100 is translated into the physical address 0x2100. There is a TLB miss and a L1 cache miss (the block on line 0 is evicted). After the lw instruction the status of the TLB and of the L1 cache is:

Valid	VPN	PFN
1	0x10	0x20
1	0x11	0x21

Index	Validity bit	Tag	Block
0	1	0x21	16 B
1	0		
...	...	...	...
15	0		

- lw \$t0, 0x1000(\$zero)

The virtual address 0x1000 is translated into the physical address 0x2000. There is a TLB hit and a L1 cache miss (the block on line 0 is evicted). After the lw instruction the status of the TLB and of the L1 cache is:

Valid	VPN	PFN
1	0x10	0x20
1	0x11	0x21

Index	Validity bit	Tag	Block
0	1	0x20	16 B
1	0		
...	...	...	...
15	0		

In summary we have: 2 TLB misses and 2 TLB hits, 1 L1 hit and 3 L1 misses.

$$AMAT = T_{L1HIT} + MR_{L1} \times \text{Miss Penalty} = 1 + \frac{3}{4} \times 50 = 38.5$$

$$EMAT = AMAT + T_{TLBHIT} + MR_{TLB} \times T_{PT} = 38.5 + 1 + \frac{1}{2} \times 20 = 49.5$$

Exercise 2

A processor has 16-bit virtual addresses and a page size of 256 B. The systems have a two-levels page table. The processor is equipped with a 2-entry fully associative TLB with LRU replacement policy. Moreover, it has a direct mapped L1 cache of 256 B with a block of 16 B. Consider the following assembly code and compute EMAT and AMAT. Assume that $T_{L1HIT} = 1$ cycle, Miss Penalty = 50 cycles, $T_{TLBHIT} = 1$ cycle, $T_{PT} = 20$ cycles (time taken by the page walk in case of TLB miss).

```
lw $t0, 0x1000($zero)
lw $t1, 0x1004($zero)
lw $t2, 0x1100($zero)
lw $t3, 0x2000($zero)
lw $t4, 0x1000($zero)
```

Consider that the page tables have been initialized by the OS with the following content:

Page Directory (level 1)

L1 index	Second-level page table
0x1	PT1
0x2	PT2

PT 1

L2 index	PFN	Valid
0x0	0x20	1
0x1	0x21	1

PT 2

L2 index	PFN	Valid
0x0	0x30	1

Solution

Virtual address split: 8 bit VPN (divided into 4 bit for L1 and 4 bit for L2) and 8 bit offset

Physical address split: 8 bit tag, 4 bit cache index, 4 bit offset

TLB and cache initially empty.

Valid	VPN	PFN
0		
0		

Index	Validity bit	Tag	Block
0	0		
1	0		
...	...	...	...
15	0		

- lw \$t0, 0x1000(\$zero)

The virtual address 0x1000 is translated into the physical address 0x2000. There is a TLB miss and a L1 cache miss. After the lw instruction the status of the TLB and of the L1 cache is:

Valid	VPN	PFN
1	0x10	0x20
0		

Index	Validity bit	Tag	Block
0	1	0x20	16 B
1	0		
...	...	...	...
15	0		

- lw \$t0, 0x1004(\$zero)

The virtual address 0x1004 is translated into the physical address 0x2004. There is a TLB hit a L1 cache hit. After the lw instruction the status of the TLB and of the L1 cache is:

Valid	VPN	PFN
1	0x10	0x20
0		

Index	Validity bit	Tag	Block
0	1	0x20	16 B
1	0		
...	...	...	...
15	0		

- lw \$t0, 0x1100(\$zero)

The virtual address 0x1100 is translated into the physical address 0x2100. There is a TLB miss and a L1 cache miss. After the lw instruction the status of the TLB and of the L1 cache is:

Valid	VPN	PFN
1	0x10	0x20
1	0x11	0x21

Index	Validity bit	Tag	Block
0	1	0x21	16 B
1	0		
...	...	...	...
15	0		

- lw \$t0, 0x2000(\$zero)

The virtual address 0x2000 is translated into the physical address 0x3000. There is a TLB miss and a L1 cache miss. After the lw instruction the status of the TLB and of the L1 cache is:

Valid	VPN	PFN
1	0x20	0x30
1	0x11	0x21

Index	Validity bit	Tag	Block
0	1	0x30	16 B
1	0		
...	...	...	...
15	0		

- lw \$t0, 0x1000(\$zero)

The virtual address 0x1000 is translated into the physical address 0x2000. There is a TLB miss and a L1 cache miss. After the lw instruction the status of the TLB and of the L1 cache is:

Valid	VPN	PFN
1	0x20	0x30
1	0x10	0x20

Index	Validity bit	Tag	Block
0	1	0x20	16 B
1	0		
...	...	...	...
15	0		

$$AMAT = T_{L1HIT} + MR_{L1} \times \text{Miss Penalty} = 1 + 4/5 \times 50 = 41$$

$$EMAT = AMAT + T_{TLBHIT} + MR_{TLB} \times T_{PT} = 41 + 1 + 4/5 \times 20 = 58$$